

Flower Classification

<https://github.com/aiyshwary/flower-classification-project>

Python

TensorFlow

Keras

OpenCV

scikit-learn

Matplotlib

Seaborn

OVERVIEW

A CNN-based image classification project that identifies five flower types (Daisy, Dandelion, Rose, Sunflower, Tulip) using the Kaggle Flowers Recognition dataset and a custom TensorFlow/Keras model pipeline.

IMPACT

Built an end-to-end notebook workflow that prepares the dataset, trains a custom CNN, and evaluates multi-class accuracy to classify flower species from images.

KEY DETAILS

- Organized the Kaggle Flowers Recognition dataset into class-specific folders for five flower categories.
- Preprocessed images by resizing and normalizing pixel values, then created train/validation splits for model evaluation.
- Implemented a custom CNN architecture in TensorFlow/Keras for 5-class image classification.
- Tracked training/validation accuracy and loss across epochs and visualized results with Matplotlib/Seaborn.
- Packaged the workflow into a reproducible Jupyter Notebook for easy experimentation and iteration.